Southern New Hampshire University

CS 255 Business Requirements Document Template

System Components and Design

Purpose

- Build an integrated cloud-based system for DriverPass to address the 65% failure rate of DMV applicants by managing online learning and on-the-road driver testing (CS 255 Dri... pp. 1, 3).

- Establish a multi-tiered management platform supporting specific feature dashboards for the Owner (Liam), IT Officer (Ian), Office Secretary, and Customers (CS 255 Dri... p. 2).

- Coordinate the operational logistics of the driving school's assets, specifically managing a fleet of 10 cars and their respective dedicated trainers (CS 255 Dri... p. 2).

System Background

- DriverPass targets a critical educational gap where students fail driving exams because they limit their preparation exclusively to studying past written tests (CS 255 Dri... p. 1).

- The company delivers a hybrid business model combining text-based online coursework, electronic practice exams, classroom training, and real-time physical driving practice (CS 255 Dri... pp. 1-2).

- The system replaces manual business tracking with automated record validation, resource coordination, and cross-platform access control (CS 255 Dri... pp. 1-2).

Objectives and Goals

- Provide automated account creation and package workflows to process client bookings efficiently over the web (CS 255 Dri... pp. 2-3).

- Synchronize student scheduling to link customer profiles cleanly with specific time blocks, vehicle assignments, and available instructors (CS 255 Dri... p. 2).

- Facilitate data-driven management by enabling seamless operational reporting, audit-trail tracking, and compliance syncs with the state DMV (CS 255 Dri... pp. 1-3).

Requirements

Nonfunctional Requirements

Performance Requirements

- The architecture must run natively inside standard web browsers over a cloud deployment to offload manual backup management and infrastructure overhead (CS 255 Dri... p. 3).

- The interface must permit instant download privileges for generated system metrics, converting report streams into external spreadsheets (e.g., Microsoft Excel) for offsite evaluation (CS 255 Dri... p. 1).

- System synchronization parameters must enforce strict online-only write privileges, ensuring data modifications can only execute when a device is connected to prevent server conflicts and data redundancy (CS 255 Dri... p. 1).

Platform Constraints

- The application interface must remain responsive to function uniformly across modern computers and desktop configurations as well as mobile smartphone screens (CS 255 Dri... p. 1).
- The backend architecture must interface with an external DMV data channel to automatically consume updates, policy notifications, and revised sample test questions (CS 255 Dri... p. 3).
- The structural schema must allow modules to be toggled off at the root administrative level, giving administrators the ability to disable specific packages from client-facing view without rewriting source code (CS 255 Dri... p. 3).

Accuracy and Precision

- Every physical lesson entity recorded must enforce strict tracking, mapping a exact user to a distinct driver, a specific 2-hour timeframe, and 1 of the 10 fleet cars (CS 255 Dri... p. 2).
- Input validation masks for scheduling and onboarding forms must prevent temporal overlap errors, blocking double-bookings of any car or trainer asset (CS 255 Dri... p. 2).
- The drop-off address field input string validation must precisely match and replicate the user-submitted pickup location address string (CS 255 Dri... p. 3).

Adaptability

- The architecture must operate adaptively to allow core administrative roles to configure and adjust packages, toggle access status flags, and update system profiles on the fly (CS 255 Dri... pp. 2-3).
- The application layers must scale seamlessly from standard administrative workflows down to independent in-vehicle updates handled by active driving trainers on mobile devices (CS 255 Dri... pp. 1, 4).
- The software must adapt instantly to DMV testing modifications by refreshing its question bank based on live data notifications pushed from external regulatory endpoints (CS 255 Dri... p. 3).

Security

- The identity management subsystem must define granular, role-based access control (RBAC) tiers separating permissions for the Owner, IT Officer, Secretary, and Students (CS 255 Dri... p. 2).
- The IT Officer account tier must maintain absolute read/write over system directory trees, enabling immediate manual account overrides, forced security blocks, and user credential wipes (CS 255 Dri... p. 2).

- The environment must feature a secure self-service portal that triggers automated password reset workflows via a user-initiated request line (CS 255 Dri... pp. 2-3).
- The platform must write continuous, non-repudiable audit logs that record the identity of any account generating, editing, or deleting a reservation block (CS 255 Dri... p. 2).

Functional Requirements

- The system shall authenticate user logins and present an interactive UI tailored to the individual's security role (Owner, IT Officer, Secretary, Student) (CS 255 Dri... p. 2).
- The system shall log student intake inputs over a secure registration form capturing first name, last name, physical address, phone number, state, credit card number, expiration date, and CVV security code (CS 255 Dri... p. 3).
- The system shall store student intake metrics including specified pickup location and ensure the drop-off location field explicitly mirrors the pickup data (CS 255 Dri... p. 3).
- The system shall enable students and the office secretary to generate, modify, and delete reservations for driving sessions online (CS 255 Dri... p. 2).
- The system shall divide scheduling blocks into fixed 2-hour increments and allocate lessons based on Package One (6 hours across 3 blocks), Package Two (8 hours plus in-person training), or Package Three (12 hours, in-person training, and online materials) (CS 255 Dri... p. 2).
- The system shall display online test progress tracking, detailing test name, timestamp, score metrics, and current evaluation status (not taken, in progress, failed, passed) (CS 255 Dri... p. 4).
- The system shall present an interactive trainer log interface displaying lesson time slots, start hours, end hours, and alphanumeric driver commentary fields (CS 255 Dri... p. 4).
- The system shall provide a master administrative control toggle to disable registrations for any service package instantly (CS 255 Dri... p. 3).
- The system shall generate printable security activity logs to trace modifications back to the responsible user account (CS 255 Dri... p. 2).

User Interface

- The system interface layout must integrate a distinct corporate branding logo block positioned at the top center of the web framework (CS 255 Dri... p. 4).
- The primary dashboard interface must render a multi-grid interface showing online test progress, account records, driving instructor commentary notes, special needs entries, and profile photo slots for both trainers and students (CS 255 Dri... p. 4).
- The platform must feature an integrated messaging panel offering separate communication channels to contact the business or reach out to individual students (CS 255 Dri... p. 4).

Assumptions

- It is assumed that customers maintain a reliable mobile cellular data connection or broadband internet line during interaction cycles with the training portal (CS 255 Dri... p. 1).
- It is assumed that the external Department of Motor Vehicles system provides a standard, reliable API endpoint capable of distributing rule notifications and sample questions (CS 255 Dri... p. 3).

- It is assumed that cloud hosting architectures handle low-level failover systems, backup redundancy, and environmental security patches natively (CS 255 Dri... p. 3).

Limitations

- The core codebase built during this deployment phase will not permit non-technical staff to manually add entirely new structural software modules or custom functional packages without an analyst or engineer (CS 255 Dri... p. 3).

- Write capabilities, data modifications, and reservation edits cannot execute in offline modes; users are strictly limited to read-only reporting when disconnected from internet access (CS 255 Dri... p. 1).

- Fleet operational logistics are restricted to the 10 vehicles owned by the business and their direct trainer alignments (CS 255 Dri... p. 2).

Gantt Chart